

Unit 6:
WAN, Network Management, and Enterprise Networks

6.1 WAN Technologies, Network Monitoring, and Troubleshooting Tools

Learning Objectives



By the end of this lesson, students will be able to:

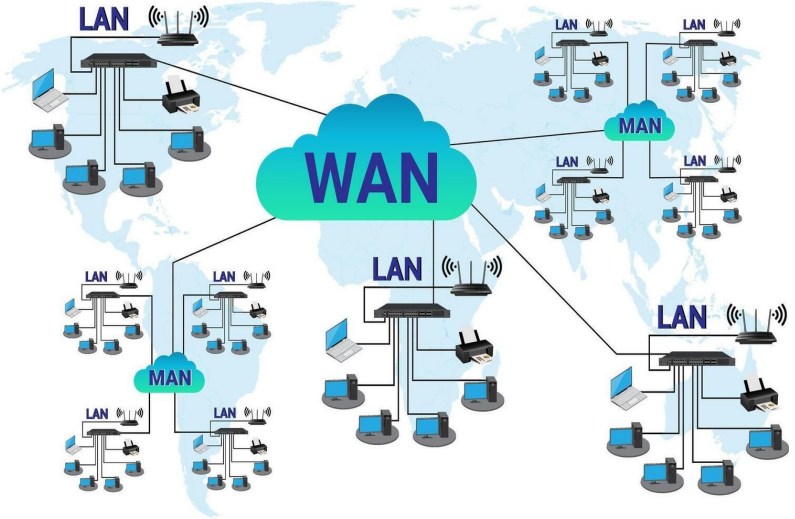
✅ Explain WAN technologies (MPLS, Metro Ethernet, PPPoE)
✅ Explain how MPLS, Metro Ethernet, and PPPoE work
✅ Understand SNMP components (polling, traps, MIBs)
✅ Describe NetFlow components and flow records
✅ Explain basic system logging concepts (syslog)
✅ Use ping, traceroute, Wireshark, and Nmap for troubleshooting

WAN Technologies

Definition: Technologies that connect networks over large geographical distances (cities, countries, continents).

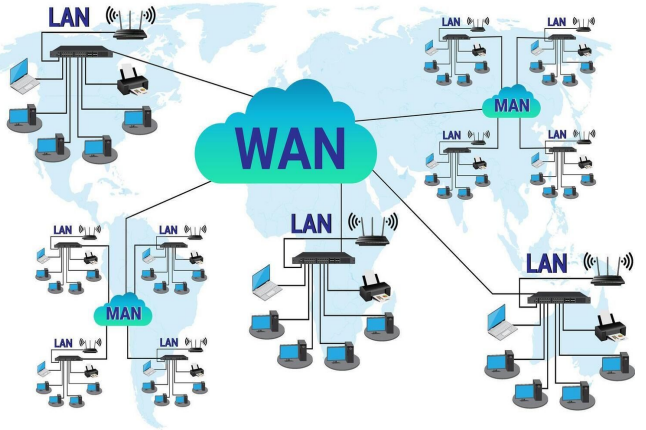
Analogy: A network of highways connecting different cities – each city is a LAN, highways are the WAN.

What they provide: Long-distance connectivity, scalability, and carrier-grade reliability.



Three Common WAN Technologies

Technology	Best for	Key feature
MPLS	Large enterprises	Label switching + VPN
Metro Ethernet	City-wide connectivity	Familiar Ethernet
PPPoE	Broadband internet	Per-user authentication



Analogy: MPLS = express lanes, Metro Ethernet = local highways, PPPoE = toll booths.

MPLS – Multiprotocol Label Switching

Definition: A switching technology that uses short labels instead of long IP addresses to forward packets faster.

How it works:
Data enters → Label assigned → Label switching occurs → Labels removed → Data delivered

Advantages:

- Faster than IP routing
- Traffic engineering (choose path based on congestion)
- Built-in VPN support (MPLS L3 VPN)

Analogy: Airport luggage tags (labels) instead of reading destination on every suitcase.

MPLS – (Use Cases)

Use Case	Description
Enterprise WAN	Connect branch offices with guaranteed SLA
Service provider backbone	Core routing for ISP networks
Layer 3 VPN	Customer sites appear on same private network
Traffic engineering	Route around congestion automatically

User experience: Employees access resources across sites as if they were local.

Cost consideration: More expensive than internet VPN, but offers SLA guarantees.

Metro Ethernet – Carrier Ethernet

Definition: Ethernet-based WAN technology used to connect sites within a metropolitan area (up to ~50km radius).

How it works:
[Office A] ===== Metro Ethernet Cloud ===== [Office B] + [Data Center]

Key Features:

- Familiar Ethernet framing (no new protocols to learn)
- High bandwidth: 10Mbps to 100Gbps
- Lower cost than traditional WAN (leased lines)
- Scalable (add bandwidth incrementally)

Analogy: One giant Ethernet switch spread across the entire city

Metro Ethernet – (Use Cases)

Use Case	Description
Campus connectivity	Connect multiple buildings on university campus
Disaster recovery	Backup link between primary and DR data centers
Cloud on-ramp	Direct connection to AWS/Azure/Google Cloud
Service provider handoff	ISP delivers internet via Metro Ethernet

Benefits over MPLS: Lower cost per Mbps, simpler configuration, easier troubleshooting.
Limitation: Metro area only – not for inter-city or inter-country.

PPPoE – PPP over Ethernet

Definition: Encapsulates PPP frames inside Ethernet frames – combines PPP's authentication with Ethernet's ubiquity.

How it works:
[Home Router] → (PPPoE tunnel) → [ISP BRAS] → Internet

Why PPOE?

- PPP gives you: username/password authentication, dynamic IP assignment, session control, encryption (optional)
- Ethernet gives you: cheap, ubiquitous, high-speed connectivity

Analogy: Showing your membership card (PPP) at a stadium gate (Ethernet).

PPPoE – (Use Cases)

When to use Metro Ethernet:

Use Case	Description
Residential broadband	DSL and fiber home internet connections
Remote access	Individual remote workers connecting to ISP
Wholesale ISP services	One ISP provides wholesale access to another
Mobile offload	Cellular carrier offloads to Wi-Fi with PPPoE

Advantages

- Per-user authentication and accounting
- Session management (ISP can see who is connected)
- Works with existing RADIUS servers

Disadvantages

- MTU issues (PPPoE adds 8 bytes, needs lower MTU)
- Slight performance overhead

Network Monitoring

Why monitor your network?

- Detect failures before users complain
- Plan capacity upgrades (when to buy more bandwidth)
- Security incident detection (unusual traffic patterns)
- Compliance reporting (prove SLA was met)
- Historical analysis (trends over time)

The three pillars of network monitoring:

Pillar	Protocol	What it tells you
Device health	SNMP	CPU, memory, interface status, temperature
Traffic flow	NetFlow	Who talked to whom, how much data
Events & logs	Syslog	What happened and when

SNMP – Simple Network Management Protocol

Definition: Protocol for monitoring/managing network devices (routers, switches, servers).

Operates at: Layer 7 (Application)

Three components:

- Manager – Central server that requests data
- Agent – Software on device that collects data
- MIB – Hierarchical database of monitorable objects (OID tree)

Analogy: SNMP is like a security guard who walks around checking on every device.

SNMP – Simple Network Management Protocol

Polling vs. Traps

Polling is when the SNMP manager actively requests data from agents at regular intervals (every 5 minutes) to collect routine statistics like CPU usage or interface traffic. Traps are the opposite – agents send unsolicited alerts to the manager immediately when an event occurs, such as a link going down. Think of polling as a teacher calling roll to see who is present, while traps are a student yelling "FIRE!" without being asked. Polling is reliable but uses steady bandwidth, whereas traps are event-driven but can be missed if the network drops the packet.

Operation	Direction	Use
Polling	Manager → Agent	Regular stats (every 5 min)
Traps	Agent → Manager	Urgent alerts (link down)

NetFlow – Flow-Based Monitoring

Definition: Collects IP traffic metadata – not full packets. Cisco protocol, now IPFIX standard.

- **What is a flow?** A unidirectional sequence of packets with the same: source IP, destination IP, source port, destination port, protocol, ToS, input interface.
- **Purpose:** Understand who is talking to whom, how much data, and when.
- **Analogy:** NetFlow is like a phone bill – shows who called whom and for how long, but not the conversation content.

Flow record contains: Timestamps, packet/byte counts, TCP flags, next-hop IP, AS numbers.

NetFlow – Flow-Based Monitoring

Component	Function	Example
Exporter	Device that samples traffic and creates flow records	Cisco router, switch, firewall
Collector	Server that receives and stores flow records	SolarWinds, ELK Stack, PRTG
Analyzer	Tool that visualizes and analyzes flow data	Same as collector (integrated)

What NetFlow does NOT give: Full packet contents, application payload, real-time 100% accuracy.

Syslog – Basic System Logging

Definition: Standard protocol (UDP 514) for sending log messages from devices to a central server.

Message format: `<priority>timestamp hostname process: message`

Example: `<34>Feb 11 14:23:15 router1 %LINK-3-UPDOWN: Interface Gig0/1, changed state to up`

Severity levels (0–7):

- 0-2 (Emergency/Critical) → Page on-call engineer
- 3 (Error) → Investigate within 24 hours
- 4-6 (Warning/Notice/Info) → Log only
- 7 (Debug) → Do not log in production

Best practice: Centralize logs, monitor 0-3 in real time, rotate logs (30-day retention), use NTP for accurate timestamps.

Analogy: Syslog = black box flight recorder for your network.

Troubleshooting Tools – Overview

Tool	Purpose	Analogy
Ping	Test reachability and RTT	Sonar ping on a submarine
Traceroute	Map path to destination	Breadcrumb trail
Wireshark	Capture and inspect packets	Security camera footage
Nmap	Discover hosts and open ports	Flashlight in a dark room

Methodology: Start with ping → if fail, traceroute → if still broken, Wireshark.

Ping, Traceroute, Wireshark & Nmap

Ping: `ping 8.8.8.8` → shows reply, RTT, packet loss.

Traceroute: `traceroute google.com` → shows each hop and per-hop latency.

Wireshark: Capture packets, use filters like `ip.addr==x` or `tcp.port==80`. Follow TCP streams to see whole conversations.

Nmap: `nmap 192.168.1.0/24` (ping sweep), `nmap -sV target` (service version), `nmap -O target` (OS detection). Only scan networks you own or have permission to scan.

Key Takeaways

- **MPLS** – Uses short labels instead of full IP addresses to forward packets faster; also supports built-in VPNs for connecting multiple offices.
- **Metro Ethernet** – Brings Ethernet technology across a whole city, offering high bandwidth (up to 100 Gbps) at lower cost than traditional leased lines.
- **PPPoE** – Combines PPP authentication (username/password) with Ethernet; this is what your home DSL or fiber router uses to log in to the ISP.
- **SNMP** – Monitors device health using polling (manager asks agents every few minutes) and traps (agents send urgent alerts); the MIB is a catalog of what can be monitored.
- **NetFlow** – Collects metadata about traffic (who talked to whom, how much data, when) – like a phone bill, not the conversation. Never captures packet contents.
- **Syslog** – Sends log messages from devices to a central server; each message has a severity level from 0 (Emergency) to 7 (Debug). Monitor levels 0-3 for immediate action.
- **Ping** – Tests if a remote host is reachable and measures round-trip time (RTT) and packet loss. Always the first troubleshooting tool.
- **Traceroute** – Shows every router hop between you and a destination, along with the delay at each hop. Helps find exactly where a connection is slow or broken.
- **Wireshark** – Captures live packets and allows deep inspection; you can filter traffic, follow entire TCP streams, and spot problems like retransmissions.
- **Nmap** – Discovers live hosts, open ports, running services, and operating systems on your network. Essential for inventory and security checks.

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Thank You!